Session 08 (AIML) -Assignment Question

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- **College Name:** Zeal Polytechnic , Narhe
- **Branch Name:** AIML

Q1.

PROGRAM:

```python
session 8.py > ...
# ---------------- Q1. Creating Sets ----------------

# a) A set with 5 integers
set1 = {10, 20, 30, 40, 50}
print("Set with 5 integers:", set1)
print("Type:", type(set1))

# b) A set with mixed data types
set2 = {100, "Python", 3.14}
print("\nMixed data type set:", set2)
print("Type:", type(set2))

# c) Empty set using correct method
empty_set = set()
print("\nEmpty set:", empty_set)
print("Type:", type(empty_set))

# d) Set from string "hello"
set3 = set("hello")
print("\nSet from string 'hello':", set3)
print("Type:", type(set3))

# Sets automatically remove duplicate values
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR>  & 'C:\Users\tanve\AppData\L
-x64\bundled\libs\debugpy\launcher' '56340' '--' 'c:\Users\
Set with 5 integers: {50, 20, 40, 10, 30}
Type: <class 'set'>

Mixed data type set: {'Python', 3.14, 100}
Type: <class 'set'>

Empty set: set()
Type: <class 'set'>

Set from string 'hello': {'o', 'e', 'h', 'l'}
Type: <class 'set'>
PS C:\Users\tanve\Desktop\ITR>
```

Q2.

PROGRAM:

session 8.py > ...

```python
# --------------- Q2. Characteristics of Sets ---------------

numbers = {10, 20, 30, 20, 40, 10, 50}

# Duplicate values are removed automatically
print("Set:", numbers)

# Sets are unordered
print("Print 1:", numbers)
print("Print 2:", numbers)

# Indexing is not allowed in sets
# print(numbers[0])   # TypeError: 'set' object is not subscriptable
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR>  c:; cd 'c:\Users\t
ions\ms-python.debugpy-2026.6.0-win32-x64\bundled\
Set: {50, 20, 40, 10, 30}
Print 1: {50, 20, 40, 10, 30}
Print 2: {50, 20, 40, 10, 30}
PS C:\Users\tanve\Desktop\ITR>
```

Q3.

PROGRAM:

```
session 8.py > ...
# ---------------- Q3. Membership Testing ----------------

# Taking 5 colors as input
colors = set()

for i in range(5):
    color = input(f"Enter color {i+1}: ")
    colors.add(color)

print("Color set:", colors)

check_color = input("Enter a color to search: ")

if check_color in colors:
    print(check_color, "is present in the set.")
else:
    print(check_color, "is not present in the set.")

if check_color not in colors:
    print(check_color, "is absent from the set.")
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR>  c:; cd 'c:\Users\tanve\Desktop
ions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy'
Enter color 1: pink
Enter color 2: blue
Enter color 3: green
Enter color 4: orange
Enter color 5: black
Color set: {'orange', 'pink', 'black', 'blue', 'green'}
Enter a color to search: pink
pink is present in the set.
PS C:\Users\tanve\Desktop\ITR>
```

Q4.

PROGRAM:

```python
# --------------- Q4. add(), remove(), discard() ---------------

fruits = {'apple', 'banana', 'mango'}

# a) Add orange
fruits.add('orange')
print("After adding orange:", fruits)

# b) Add banana again (no duplicate added)
fruits.add('banana')
print("After adding banana again:", fruits)

# c) Remove mango
fruits.remove('mango')
print("After removing mango:", fruits)

# d) Discard grape (no error if element not found)
fruits.discard('grape')
print("After discarding grape:", fruits)
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR>  c:; cd 'c:\Users\tanve\Desktop\ITR'
ions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\laun
After adding orange: {'banana', 'orange', 'mango', 'apple'}
After adding banana again: {'banana', 'orange', 'mango', 'apple'}
After removing mango: {'banana', 'orange', 'apple'}
After discarding grape: {'banana', 'orange', 'apple'}
PS C:\Users\tanve\Desktop\ITR>
```

Q5.

PROGRAM:

```
session 8.py > ...
# ---------------- Q5. pop() and clear() ----------------

s = {100, 200, 300, 400, 500}

# Remove and print one element
removed_element = s.pop()
print("Removed element:", removed_element)

# Print set after pop()
print("Set after pop:", s)

# Clear entire set
s.clear()

# Print final empty set
print("Final set:", s)
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR>  c:; cd 'c:\Users
ions\ms-python.debugpy-2026.6.0-win32-x64\bundle
Removed element: 400
Set after pop: {100, 500, 200, 300}
Final set: set()
PS C:\Users\tanve\Desktop\ITR> 
```

Q6.

PROGRAM:

```
session 8.py > ...
# Create two sets
set1 = {1, 2, 3}
set2 = {3, 4, 5, 6}

# Make a copy of set1
copy_set = set1.copy()

# Update set1 with elements from set2
set1.update(set2)

# Print original copy and updated set
print("Original copy:", copy_set)
print("Updated set1:", set1)
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR>  c:; cd 'c:\User
ions\ms-python.debugpy-2026.6.0-win32-x64\bundl
Original copy: {1, 2, 3}
Updated set1: {1, 2, 3, 4, 5, 6}
PS C:\Users\tanve\Desktop\ITR>
```

Q7.

PROGRAM:

session 8.py > ...

```python
A = {1, 2, 3, 4, 5}
B = {4, 5, 6, 7, 8}

# Union using union() method
print("Union using union():", A.union(B))

# Union using | operator
print("Union using | :", A | B)

# Intersection using intersection() method
print("Intersection using intersection():", A.intersection(B))

# Intersection using & operator
print("Intersection using & :", A & B)
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR>  c:; cd 'c:\Users\tanve\De
ions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\de
Union using union(): {1, 2, 3, 4, 5, 6, 7, 8}
Union using | : {1, 2, 3, 4, 5, 6, 7, 8}
Intersection using intersection(): {4, 5}
Intersection using & : {4, 5}
PS C:\Users\tanve\Desktop\ITR>
```

Q8.

PROGRAM:

```python
session 8.py > ...
# Create an empty set
numbers = set()

# Take 6 numbers as input
for i in range(6):
    num = int(input("Enter a number: "))
    numbers.add(num)

print("Set after input:", numbers)

# Add two more numbers
numbers.add(int(input("Enter another number: ")))
numbers.add(int(input("Enter one more number: ")))

# Remove one number safely
remove_num = int(input("Enter a number to remove: "))
numbers.discard(remove_num)

# Print final set and its length
print("Final set:", numbers)
print("Length of set:", len(numbers))
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR>  c:; cd 'c:\Users\tanv
ions\ms-python.debugpy-2026.6.0-win32-x64\bundled\lib
Enter a number: 26
Enter a number: 25
Enter a number: 11
Enter a number: 05
Enter a number: 07
Enter a number: 2023
Set after input: {5, 7, 2023, 11, 25, 26}
Enter another number: 8
Enter one more number: 9
Enter a number to remove: 7
Final set: {5, 2023, 8, 9, 11, 25, 26}
Length of set: 7
PS C:\Users\tanve\Desktop\ITR>
```

Q9.

PROGRAM:

```
session 8.py > ...
scores = [85, 92, 78, 92, 85, 88, 95, 78]

# Convert list to set
unique_scores = set(scores)

# Print unique scores
print("Unique scores:", unique_scores)

# Convert set to sorted list
sorted_scores = sorted(unique_scores)
print("Sorted unique scores:", sorted_scores)

# Total number of unique scores
print("Total unique scores:", len(unique_scores))
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR>  c:; cd 'c:\Users\tanv
ions\ms-python.debugpy-2026.6.0-win32-x64\bundled\lib
Unique scores: {78, 85, 88, 92, 95}
Sorted unique scores: [78, 85, 88, 92, 95]
Total unique scores: 5
PS C:\Users\tanve\Desktop\ITR>
```

Q10.

PROGRAM:

session 8.py > ...

```python
items = set()

while True:
    print("\n--- MENU ---")
    print("1. Add item")
    print("2. Remove item")
    print("3. Show all unique items")
    print("4. Check if item exists")
    print("5. Clear all items")
    print("6. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        item = input("Enter item to add: ")
        items.add(item)
        print("Item added.")

    elif choice == 2:
        item = input("Enter item to remove: ")
        items.discard(item)
        print("Item removed (if present).")

    elif choice == 3:
        print("Unique items:", items)

    elif choice == 4:
        item = input("Enter item to check: ")
        if item in items:
            print("Item exists.")
        else:
            print("Item does not exist.")

    elif choice == 5:
        items.clear()
        print("All items cleared.")

    elif choice == 6:
        print("Exiting program...")
        break

    else:
        print("Invalid choice!")
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR>  & 'C:\Users\tanve\AppData\Local\Python\pytho
-x64\bundled\libs\debugpy\launcher' '59941' '--' 'c:\Users\tanve\Desktop\ITR

--- MENU ---
1. Add item
2. Remove item
3. Show all unique items
4. Check if item exists
5. Clear all items
6. Exit
Enter your choice: 4
Enter item to check: mango
Item does not exist.

--- MENU ---
1. Add item
2. Remove item
3. Show all unique items
4. Check if item exists
5. Clear all items
6. Exit
Enter your choice: 2
Enter item to remove: banana
Item removed (if present).

--- MENU ---
1. Add item
2. Remove item
3. Show all unique items
4. Check if item exists
5. Clear all items
6. Exit
Enter your choice: 5
All items cleared.

--- MENU ---
1. Add item
2. Remove item
3. Show all unique items
4. Check if item exists
5. Clear all items
6. Exit
Enter your choice: 6
Exiting program...
```